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MILITARY SPECIFICATION

OVENS, BAKING AND ROASTING, FORCED CONVECTION, ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one- and two-compartment, baking and roasting, electric, forced convection ovens.

1.2 Classification. Ovens shall be of the following types and sizes as specified (see 6.2).

Type I	-	One oven compartment
Type II	-	Two oven compartment
Size 1	-	Standard depth oven cavity
Size 2	-	Extra deep oven cavity

1.2.1 Military part number. The military part number shall consist of the letter M, the basic number of the specification and the appropriate numbering sequence, as specified in 6.4.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7310

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SPECIFICATIONS

FEDERAL

QQ-S-698 - Steel, Sheet and Strip, Low Carbon
RR-P-58 - Pans, Bread

MILITARY

MIL-B-2316 - Bakery Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-167/1 - Mechanical Vibration of Shipboard Equipment Type I - Environmental and Type II - Internally Excited
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip

A 176 - Stainless and Heat-Resisting Chromium Steel Plate Sheet, and Strip

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A 463 - Steel Sheet, Cold-Rolled, Aluminum-Coated Type 1 and Type 2

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 4 - Commercial Cooking and Warming Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Road, P.O. Box 1468, Ann Arbor, MI 48106.)

UNDERWRITERS LABORATORIES, INC. (UL)

Standard No. 197 - Commercial Electric Cooking Appliances

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard product. The oven shall, as a minimum, be in accordance with the requirements of this document and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this document. Standard or modified commercial products furnished in accordance with this document shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

3.2 Codes and standards. The convection ovens shall comply with the applicable requirements of NSF Standard No. 4 and UL Standard No. 197. Prior to the approval of the initial production lot, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that these requirements have been met (see 4.3).

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3.3 Materials (see 6.3) and components. Materials and components shall be as specified herein. Materials and components not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial convection oven, provided the completed item complies with all provisions of this document.

3.3.1 Steel.

3.3.1.1 Stainless and heat-resisting steel. The stainless and heat-resisting steel shall conform to any type in the 300 series of ASTM A 167; or any type in the 400 series of ASTM A 176.

3.3.1.2 Carbon steel. Carbon steel shall be commercial quality cold rolled (CRCQ) conforming to QQ-S-698.

3.3.1.3 Aluminum-coated steel. The aluminum-coated steel shall conform to type I of ASTM A 463.

3.4 Design. Convection ovens shall be designed for baking and roasting and shall incorporate a two speed oven blower to provide for circulation of forced, heated, air throughout the oven compartment. Type I and II ovens shall have one oven compartment and two oven compartments, respectively. The type I oven shall consist essentially of a body, oven compartment linings, doors with handle(s), heat-resisting glass viewing window(s), shelves, shelf supports, heating units, two speed oven blower, temperature control, timer, oven light, power on-off switch, blower speed selector switch (low and high), leg assembly, and shall be supplied with an adjustable air baffle to control air to the oven compartment and a vent remotely controlled from a control panel to release moisture from the oven cavity. The type II oven shall consist of two single ovens, as specified above, stacked and secured. Each compartment of the type II oven shall be capable of being operated independently and shall be individually controlled by a thermostat. Size 1 ovens shall be provided with standard depth oven cavity and size 2 ovens shall be provided with an extra deep oven cavity (see table I).

3.4.1 Dimensions. Ovens shall be within the dimensional limits shown in table I.

TABLE I. Dimensions (inches)

Measurements	Size 1		Size 2	
	Minimum	Maximum	Minimum	Maximum
Width overall	33	40	33	40
Depth overall	32	39	38	46
Height 1/ overall (excluding vent):				
Type I	54	65	54	69
Type II	57	78	52	78

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TABLE I. Dimensions (inches) (cont'd)

Measurements	Size 1 Minimum	Size 1 Maximum	Size 2 Minimum	Size 2 Maximum
Height 1/ to bottom deck:				
Type I	26	38	26	38
Type II	11	17	11	17
Height 1/ to top deck (type II)	30	51	30	51
Compartment (inside):				
Width	26	30	26	30
Depth	19	25	27	38
Height	19	22	19	22

1/ Height from floor, including supporting legs and base.

2/ Compartment configuration shall be so arranged that each shelf/rack of size 1 ovens will accommodate a nominal 18-inch by 26-inch sheet pan or roasting pan, and each shelf/rack of size 2 ovens will accommodate two 5-strap bread pans, conforming to style 4, size 2 of RR-P-58. Pans specified shall not interfere with oven interior and contents of the loaded pans shall not contact oven surfaces, shelves, or racks during roasting or baking operations.

3.5 Construction. The components for type I and II ovens shall be fabricated of stainless steel, carbon steel, or aluminum-coated steel conforming to 3.3.1.1, 3.3.1.2, and 3.3.1.3, respectively. The type I and II ovens shall support a uniformly distributed load of 500 pounds without any visible permanent deformation when tested as specified in 4.4.1.

3.5.1 Body. The body shall be of a frame or shell construction. When frame construction is used, the frame shall consist of a base frame, top frame, and vertical angles welded together or fastened together by any other means used by the manufacturer to form a rigid assembly, or exterior sheets shall be welded to the frame enclosing it therein. When shell construction is used, the outer shell shall be stiffened and reinforced. The front of the type I and II ovens shall be provided with means for mounting the controls. When the exterior housing parts are assembled, there shall be no cracks or crevices which might retain food particles thus encouraging insect infestation. When ovens are stacked (type II) closure pieces shall be provided which will effectively seal against entrance of foreign bodies. Stacking of ovens shall not produce inaccessible voids. Means shall be provided to prevent closure pieces from being displaced during normal usage.

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3.5.1.1 Type I. The exterior surfaces of the type I oven, except the bottom, shall be fabricated of stainless steel. The bottom surface shall be fabricated of carbon steel.

3.5.1.2 Type II. The exterior surfaces of the type II oven which are the oven front, sides, back, and top shall be fabricated of stainless steel. The bottom and top surfaces of the lower oven, and the bottom surface of the upper oven may be fabricated from carbon steel provided the remaining exterior surfaces are finished with stainless steel.

3.5.2 Oven compartment linings and heater supports. Each oven baking compartment shall be provided with oven compartment linings and heater supports fabricated from stainless steel specified in 3.3.1.1, aluminum-coated steel specified in 3.3.1.3 or porcelain enamel over carbon steel specified in 3.3.1.2. Removable oven liner panels shall be provided and shall be fabricated from stainless steel specified in 3.3.1.1. The removable oven liner panels shall consist essentially of top, right-hand, left-hand and bottom panels. The oven compartment linings shall be provided without openings and shall be reinforced.

3.5.2.1 Vent and air baffle. The air baffle shall be fabricated from stainless steel specified in 3.3.1.1. The vent shall be fabricated from the same material as the oven compartment lining.

3.5.3 Doors. The doors shall be of the vertical split, side opening type. Vertically split doors when in the fully open position shall be flush with the oven interior sidewalls and shall not obstruct the removal of the shelves and shelf supports. Doors shall be equipped with handles. When one handle is provided on an oven with two doors, both doors shall operate simultaneously when opening and closing. Heat-resisting viewing window(s) shall be located in the doors to allow viewing of seventy-five percent of shelves when doors are closed. When specified (see 6.2), doors shall be without windows. Each oven shall be provided with a door control switch and a latch, clip, or other means to hold the doors closed. The oven cavity door interior shall be fabricated from the same material used for the oven compartment linings (see 3.5.2). The door control switch shall function when the oven is in operation and shall de-energize the oven blower circuit whenever the doors are opened, and energize the oven blower circuit whenever the doors are closed. If oven doors do not open or close simultaneously, each door shall have an oven blower circuit control switch. Oven doors, when closed, shall visibly seal the oven compartment.

3.5.4 Shelves. Oven compartments shall be equipped with a minimum of five equally spaced removable shelves supported by removable shelf supports. Shelves shall be fabricated of steel wire rod, steel frame with steel slats, or expanded steel. Each shelf shall be capable of supporting a load of 20 pounds per square foot without more than 1/8 inch deflection and without permanent deformation when tested as specified in 4.4.2. Each shelf shall not jam or buckle when placed into, or removed from, the shelf supports. The shelves and shelf supports shall be stainless steel, or chromium or nickel plated carbon steel.

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3.5.5 Insulation. Insulation for all ovens shall be of the blanket or block type installed in accordance with the manufacturer's commercial practice. Loose fill type insulation shall not be used.

3.5.6 Electrical characteristics. Type I and type II ovens shall have a total input rating of not less than 10.4 kilowatts (kw) per compartment. Ovens shall be equipped for operation on a 208 volt, 60 Hz, 3-phase system. When specified (see 6.2), for use in foreign locations, ovens shall be equipped for operation on a 220 volt, 50 Hz, single phase system; or 220/308 volt, 50 Hz, three phase 4-wire system.

3.5.6.1 Controls. All controls, switches, and indicator light(s) shall be located in a control panel at the front of the oven.

3.5.6.2 Heating units. Each oven shall be provided with heating units located to provide even heat distribution throughout the oven compartment. The units shall be fully protected and removable from the oven front. The heating units shall be energized by heat control switches.

3.5.6.3 Oven blower. A two speed blower shall be provided. The oven blower shall normally provide even circulation and heat distribution throughout the oven compartment when tested as specified in 4.4.6. The oven blower and heating units shall be interconnected so that the oven blower operates whenever the heating unit circuit is energized and the door(s) are closed. A separate oven blower control switch may be provided for cooling. The oven blower motor shall be of the capacitor or split phase type. The oven blower shall be adequate to sustain continuous operation at high and low oven blower motor speed settings. Motor speed on the high speed setting shall be nominal 1725 rpm and the low speed setting shall be between 750 and 1125 rpm. Switching from one speed to the other shall be accomplished without damage to the motor or to the motor electrical system. A protective device shall be provided over the blower to restrict a roasting pan handle, when in the raised position, from hitting the blower. Oven blower wheel shall be fabricated of stainless steel as specified in 3.3.1.1 or other material suitable for the purpose intended. If stainless steel is not used, the blower wheel shall be permanently coated with a material to prohibit corrosion.

3.5.6.4 Oven light. Oven light shall be used to illuminate the oven interior. The light(s) shall be recessed or protected with a heat-resistant window(s). The oven light(s) may be actuated by a manual switch, or actuated automatically, when the door is opened by a door switch. The light and protective window shall be accessible for light replacement and window cleaning.

3.5.6.5 Indicator light. All ovens shall be furnished with a separate indicator light that glows whenever the heating units and convection blower electrical system is energized or whenever the thermostat is in the "ON" position.

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3.5.6.6 Operation. The oven controls and electrical components shall operate through a complete cycle without malfunction and controls shall not bind or show looseness when tested as specified in 4.4.3.

3.5.6.7 Contactor. A magnetic contactor suitable for the rating and specified electrical characteristics of the oven shall be provided and shall be located in the control panel enclosure housing.

3.5.7 Temperature control. Each oven compartment shall be equipped with an automatic adjustable thermostat to provide automatic heat control of the oven compartment and permit cooking operations to be accomplished by maintained temperature. The thermostat shall be capable of maintaining any oven temperature within the range of 250°F to at least 400°F, plus or minus 10°F of the mean temperature setting when tested as specified in 4.4.4. The thermostat actuating components shall be protected against moisture, grease, and direct heat radiation. An "OFF" position shall be marked on the thermostat dial or a separate "ON-OFF" switch shall be provided. Temperature control dial shall have a range of 200° to at least 450°F.

3.5.7.1 Preheat time. The oven temperature shall rise to 400°F within 20 minutes in an ambient temperature of 75°F, plus or minus 5°F, when tested as specified in 4.4.5.

3.5.8 Oven time control. The oven shall be provided with a 5-hour oven timer installed in the control panel or installed in the oven front. The oven time control shall be equipped with an alarm signal which will sound with an audible ring at the completion of the preset time cycle.

3.5.9 Terminal board or box. A terminal board or box shall be furnished within the oven enclosure to provide for permanent connection to the installation wiring system. The terminal board or box shall have a grounding terminal with connection to the oven outer body.

3.6 Oven legs. The bottom of the ovens shall have four removable legs attached directly to the bottom of the oven or to a supporting base. Legs and supporting base shall be fabricated of carbon steel specified in 3.3.1.2 or of stainless steel specified in 3.3.1.1. When legs are attached directly to the oven bottom they shall be reinforced. Legs shall be removable and shall be provided with devices for leveling the oven(s) except for Naval shipboard use (see 3.9.3). When a supporting base is furnished, its extremities shall not protrude beyond the extremities of the oven exterior. The supporting base shall be reinforced and be supplied with a shelf (type I only).

3.7 Marking.

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3.7.1 Data-name plates. Each oven shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for; (a) Methods of applying, (b) Identification tags, (c) Information not required and (d) Optional marking information shall not apply. The data plates shall be made of minimum 20-gauge corrosion-resisting metal and attached to the oven by rivets, screws or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved, or applied by photosensitive means:

National stock number
Procurement instrument identification number
Specification data
Manufacturer's name, address and telephone number
Supplier's name, address and telephone number (List only if different from manufacturer)
Manufacturer's model number
Government approved manual number (see 6.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the item.

3.7.2 Users' instruction plate. Each oven shall have a users' instruction plate, permanently fastened to the outside front of the oven. The plate shall be constructed of corrosion-resistant metal having a minimum thickness of 0.0239 inch, and shall have the following information etched or stamped thereon in legible characters. The words "CONVECTION OVEN" shall be in characters not less than 3/16 inch high. Each oven control panel shall be clearly and permanently marked with identification of each control and all necessary instructions pertaining to the control. Marking shall be any method which will permanently resist obliteration by mild, abrasive cleaning compounds used in normal daily cleaning.

CONVECTION OVEN

USERS' INSTRUCTIONS

1. Turn heating unit switch(es) on and adjust thermostat to desired oven temperature.

2. Do not overload oven compartment.

3. Load shelves evenly, making sure food does not contact oven linings.

4. Clean oven daily. Clean oven exterior and interior with damp cloth or mild detergent and wipe dry. Scrape all carbonized matter from door, door frame, and outside lower lip of door using a scraper. Clean shelves daily or after each use if necessary.

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3.8 Repair and maintenance. Provisions shall be made to allow ready adjustment, service, and replacement of electrical power system components and operational electrical accessories and components, without disturbing or disassembly of any other major component or equipment.

3.9 Navy shipboard installation. For Navy shipboard installation (see 6.2), the following requirements (3.9.1 through 3.9.12) apply:

3.9.1 Removable oven lining panels. Removable oven lining panels consisting of top, right-hand, left-hand and bottom shall be fabricated from stainless steel specified in 3.3.1.1 and shall be used in conjunction with the oven chamber linings specified in 3.5.2. All oven lining panels shall be removable preferably without the use of tools or may be removable with the use of simple tools. Simple tools are defined as common or crosspoint screwdrivers, pliers or adjustable wrench.

3.9.2 Oven door latch. Oven doors (see 3.5.3) shall be equipped with secondary exterior mechanical latches and the oven cavity door interiors shall be fabricated from stainless steel specified in 3.3.1.1. Each vertically split, side opening type door shall be supplied with a hold-open device or latch which will hold the doors in a wide open position as specified in 3.5.3.

3.9.3 Legs. Ovens shall be provided with four removable legs suitable for bolting the ovens to the deck. Legs for type I ovens shall be 27 ± 1 inches long, and for type II ovens 8-1/2 ± 1 inches long. Legs shall be fabricated from 14 gage (minimum) carbon steel specified in 3.3.1.2 or of stainless steel specified in 3.3.1.1 and shall be adequately reinforced. The bottom flange of each leg shall be provided with clearance holes for hold down bolts. Each hold-down bolt shall be accessible from the front for mounting and dismounting the oven. When legs are attached directly to the oven bottom, they shall attach to a structural member or suitable reinforcing plate. Attaching holes in the oven bottom may be tapped (minimum depth one bolt diameter plus one thread) or capture nuts may be utilized.

3.9.4 Oven structural strength. The oven shall not deviate or rack from perpendicularity to the deck by more than 1/8 inch when the deck is inclined at an angle of 15 degrees in each of two planes, one normal to the oven front and one to the oven side when tested as specified in 4.4.1.1.

3.9.5 Electrical requirements. Ovens shall be equipped for operation on 440 volts, 3-phase, or 230 volts direct current system with 115 volts for controls and motor operation, as specified (see 6.2). Provision for grounding shall be included for each oven.

3.9.5.1 Structural. Electrical components of considerable mass, such as contactors, heating elements, terminal blocks, fuse blocks, switches and other devices shall be securely attached to suitable plates, brackets, etc., to prevent displacement under normal ships motion.

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3.9.5.2 Heating elements. The ends of heating elements shall be effectively sealed and protected to prevent the entrance of moisture. Insulation resistance at normal operating temperature shall be not less than 1 megohm. At room temperature, the insulation resistance shall be not less than 2 megohms. Heating elements shall be entirely removable from within the oven cavity. Leads requiring disconnection from terminal boards or strips outside the oven cavity will not be permitted.

3.9.5.3 Separate control box. A separate control box may be provided for 230 V DC system. The line terminal locks and electric accessories such as cutouts with fuses, contactors, and control transformer (when furnished) shall be suitably mounted and also enclosed within the control box. The control box shall consist of a body and removable service cover designed for overhead or bulk-head mounting. The control box shall be fabricated from stainless steel specified in 3.3.1.1, not less than 0.059 inch thick. The control box shall be arranged for normal mounting and shall not be wider than the width of the oven body. Four tapped holes located in the control box back and top shall be provided for attachment.

3.9.6 Removable shelves. Oven compartment(s) shall be provided with one shelf specified in 3.5.4 for each shelf support position.

3.9.7 Servicing. Ovens shall be constructed so that all parts are readily accessible and normal servicing can be effected from the front of the oven.

3.9.8. Oven cavity light. The oven cavity shall be provided with a light as specified in 3.5.6.4.

3.9.9 Hatchability. The oven shall be capable of passing through a hatch or doorway measuring 26 inches by 66 inches. For other dimensional limitations see 3.4.1.

3.9.10 Electromagnetic compatibility. When specified for shipboard use (see 6.2), the units shall be designed and equipped for electromagnetic compatibility in accordance with the requirements of MIL-STD-461, class A4 for surface ships and class A5 for submarines. The equipment shall meet the emission and susceptibility requirements for CE01, CE03 and RE02 (see 4.4.7).

3.9.11 Environmental suitability. When specified for shipboard use (see 6.2), ovens shall be capable of withstanding ships vibration. Controls, switches, moving parts and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage when tested as specified in 4.4.8.

3.9.12 Human factors criteria (for Navy shipboard use). Human factors engineering criteria, principles and practices, as defined in MIL-STD-1472, shall be incorporated into the design of the ovens. The equipment shall be designed such that all maintenance and operational activities shall permit safe and efficient performance by the 5th percentile female to the 95th percentile

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male as defined in MIL-STD-1472. All controls, switches and indicator lights shall be selected and integrated into the design of the ovens so as to meet the applicable requirements of MIL-STD-1472 and they shall be clearly and appropriately labelled to identify function. The clearance or free area required around an item shall permit an individual with applicable 5th to 95th percentile body dimensions and physical capabilities, as defined in MIL-STD-1472, to safely operate, maintain, remove or replace that item. When establishing accessibility requirements, both physical and visual access must be provided, along with access for any tools or test equipment. When inspecting for defects and performing tests, the equipment shall adhere to the human factors engineering considerations listed herein.

3.10 Finish. Ovens shall be smooth on all surfaces and edges and shall be free from discoloration or stains. Oven compartments shall be finished in accordance with the manufacturer's standard practice for stainless steel, aluminum-coated steel, or porcelain enamel on carbon steel.

3.11 Workmanship. All components and assemblies of the oven shall be free from dirt and other harmful extraneous material,, burrs, slivers, rough die, tool and grind marks, dents, and cracks. Castings, molded parts, and stampings, if used, shall be free of sand, fins, pits, blowholes, and sprues. External surfaces shall be free from sharp edges and sharp corners, except when corners are required. Electrical wires shall be secured so that no interference exists with servicing or operation of mechanical or electrical components.

3.11.1 Metal fabrication. Metal used in fabrication of the oven shall provide for original quality surface finish and shall be free from kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major nature shall be made with dies on fixtures to insure uniformity of size and shape.

3.11.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts, and incomplete fusion. All scale and flux shall be removed from the finish weld area.

3.11.3 Fastening devices. Threaded fasteners and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked, or stripped and shall be securely torqued. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole. Captive nuts, threaded fasteners or other securing devices shall be provided in restrictive areas where nuts and mating screws are not both accessible for tightening. Capture devices shall also be used in areas where attachment is made to other than structural members where fasteners cannot be tightened after removal and replacement, i.e., where self-tapping screws are installed to secure access panels to thin gauge sheet metal.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 In-process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the affected items and the lot in-process:

a. Carbon steel provided for porcelain enamel coating when applicable (see 3.5.2).

b. Insulation is of the specified type (see 3.5.5).

c. Attachment of legs to oven bottom accomplished as specified (see 3.6; see 3.9.3 when applicable).

d. Wires secured to prevent interference with servicing and operation (see 3.11).

e. Captive nuts, threaded fasteners or other securing devices provided in restrictive areas or in access panel mounting areas (see 3.11.3).

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4.2.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of end items. the sample unit shall be one end item. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified		X
	Enameled surfaces blistered, peeled, chipped, or area of no film.		X
	Evidence of discoloration or stains.	X	
Construction and workmanship (general)	Component missing.	X	
	Any burr, sliver, sharp edge, or rough die, tool, gouge, trim, or grind marks.	X	
	Any component fractured, buckled, kinked, bent, dented, punctured, or malformed.	X	
Body	Not constructed as specified.	X	
	Not fabricated of material specified.	X	
Oven compartment	Evidence of open joints in linings.	X	
	Linings not reinforced or supported.	X	
Doors (as applicable)	Doors without windows when required.	X	
	No door control switch.	X	
	Not full width of the oven interior.	X	
	Door windows do not permit viewing of 75 percent of shelves.	X	
	Door handle missing.	X	
	Doors when fully open obstruct removal of shelf or shelf supports.	X	
	Not provided with secondary mechanical door latch (shipboard use).	X	
	Vertically split side opening doors, not provided with hold open device or latch (shipboard use).	X	
	Doors, when closed, do not seal oven compartment	X	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Shelf and shelf support (as applicable)	Not removable.	X	
	Not proper number (i.e. less than five).	X	
	Shelf support not easily removable.	X	
Vent	Vent or adjustable air baffle missing.	X	
Heating unit	Not removable from front of oven.	X	
	Not protected or supported.	X	
	Not removable from oven cavity.	X	
Oven blower	Oven blower does not operate when heating units are energized.	X	
	Oven blower not controlled by door switch (i.e., during cooking cycle, does not shut off when door is open or turn on when door is closed).	X	
	Not located to prevent heat loss.	X	
	Not accessible for cleaning or servicing.	X	
	Oven blower wheel not fabricated of stainless steel or coated with permanent anti-corrosion material.	X	
Lights	Not accessible for replacing.	X	
	Manual light control missing.	X	
Interior lighting	Not recessed or protected in compartment.	X	
	Not located to provide maximum illumination.	X	
Indicator light	Not located in front panel.	X	
	Does not indicate when component is energized.	X	
Thermostat	Improper range.	X	
	Not protected from direct heat.	X	
Timer	Not provided with an alarm signal.	X	
Terminal board or strips, as applicable	Heating element leads outside oven cavity.	X	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Base assembly or support legs (as applicable)	Base assembly extends beyond oven sides or front.		X
	Shelf missing from base assembly.	X	
	Legs not adjustable or removable.	X	
	Leveling devices missing.	X	
	Legs not reinforced.	X	
	Legs not suitable for bolting ovens to the deck (shipboard use).	X	
Welding (when applicable)	Not free of oxide, scale, paint, or grease.	X	
	Not continuous, sound, smooth, or free from porosity, cracks, incomplete fusion, or deformation of material.	X	
	Flux (when used) not removed.		X
	Leg brackets not provided (shipboard use).	X	
Servicing (shipboard use)	Not readily accessible from front.	X	
Bolts, nuts, screws, studs rivets and other types of threaded fasteners (as applicable)	Missing, broken, stripped, or fractured.	X	
	Loose.		X
	Rivets not fully peened or upset.		X
	Rivets head(s) not in full contact with riveted members.		X
Marking	Missing, incomplete or not legible		X
Instructions	Not securely attached to oven.		X
Data-name plates	Omitted or not as specified		X
	Information incomplete or not legible		X
	Not located so as to be readily visible to operator		X

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4.2.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table I. Any dimension that is not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of end items. The sample unit shall be one end item. the inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.5 End item testing.

4.2.5.1 Representative item testing. One representative end item from the lot shall be tested as specified in 4.4.4 through 4.4.6 (and 4.4.1.1, 4.4.7, and 4.4.8 when Naval shipboard installation is applicable). Any test failure shall be cause for rejection of the lot.

4.2.5.2 Operational testing. Every end item shall be tested as specified in 4.4.3. Any end item exhibiting one or more test failures shall be rejected.

4.2.5.3 Strength testing. The end items shall be tested for strength as specified in 4.4.1 and 4.4.2. The lot size shall be expressed in units of end items. The sample unit shall be one end item. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for test failures.

4.2.6 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of MIL-B-2316.

4.3 Certification compliance examination. Certificates of compliance certified test reports, or listing marks for codes and standards that are submitted as proof of compliance with 3.2 shall be examined and validated (see 4.3.1 and 4.3.2).

4.3.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

a. A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished convection oven.

b. A certification for the convection ovens issued by NSF under their special one-time contract evaluation/certification service.

c. A test report, acceptable to the contracting officer with the advice of the Army Surgeon General, from an independent testing laboratory indicating that the items have been tested and conform to the specified NSF Standard.

4.3.2 UL. Acceptable evidence of meeting the applicable requirements of UL Standard No. 197 shall be a UL listing mark or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the ovens have been tested and conform to the applicable requirements of UL Standard No. 197.

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4.4 Methods of inspection.

4.4.1 Oven structural strength. The top of the assembled oven shall be subjected to a uniformly distributed load of 500 pounds. After 10 minutes the load shall be removed and the oven examined to determine compliance with the requirements specified in 3.5. Inability of the oven to meet the oven structural strength requirements specified shall constitute failure of this test.

4.4.1.1 Inclined and structural strength (shipboard use ovens only). After the test in 4.4.1 is completed remove the load from the top of the assembled oven and subject each oven cavity to a uniform load of 100 pounds. Position the oven with the base set at an angle of 15 degrees, then operate the oven for 30 seconds at each side of the vertical in each of two vertical planes at right angles to each other. At each of these positions observe the oven for compliance with 3.9.4. Any noncompliance with the requirements specified shall constitute failure of this test.

4.4.2 Shelf structural strength test. The shelf shall be held in its normal position in the oven and subjected to a uniformly distributed load of 20 pounds per square foot. After 2 minutes, the load shall be removed and the shelf examined to determine compliance with the load requirements specified in 3.5.4. Inability to meet the shelf load requirements specified shall constitute failure of this test.

4.4.3 Operational test. Each oven section shall be connected to a rated power source, and operated without load. During this test run, the test operator shall make an examination of all equipment and controls to determine that there is no malfunction. All manually operated controls shall be operated through a complete cycle to determine compliance with 3.5.6.6. Any nonconformance with the operational requirements specified shall constitute failure of this test.

4.4.4 Oven thermostat accuracy test. Each oven compartment shall be energized; the thermostat shall be successively set at 250°F, 300°F, and 400°F and the compartment temperature shall be measured to determine compliance with 3.5.7. Inability to meet the thermostat accuracy requirements specified shall constitute failure of this test.

4.4.5 Preheat time with blower energized test. The oven shall be at temperature of 75°F plus or minus 5°F with the door closed before starting the test. The temperature control shall be set at 400°F and the oven and convection blower energized. The temperature shall be read at 1 minute intervals. The time required to reach an oven temperature of 400°F shall be recorded to determine compliance with 3.5.7.1. Any noncompliance with the requirements specified shall constitute failure of this test.

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4.4.6 Circulation and heat distribution test. These tests shall be performed at one thermostat setting and after temperature with blower energized has stabilized. Thermocouples shall be located on the top, middle, and bottom shelves. The thermocouples shall measure heat distribution of the oven and shall be located on each shelf in the following manner: Lines shall be drawn on each shelf; 4 inches in and parallel to the front, back, and each side; mid-point to mid-point of each side; and mid-point of the back to mid-point of front. A thermocouple shall be located at each line intersection for a total of nine per shelf. All temperatures shall be read on a recording potentiometer. Use of deflection type instruments shall not be considered satisfactory. Mean temperature readings on each shelf, at one thermostat setting, shall not vary more than 25 degrees from each other. This test shall be conducted in conjunction with 4.4.5 to determine compliance with 3.5.6.2 and 3.5.6.3. Any nonconformance with the circulation and heat distribution requirements specified shall constitute failure of this test.

4.4.7 Electromagnetic interference test. When required for shipboard use, the unit shall be tested by the contractor in accordance with test methods CE01, CE03, and RE02 of MIL-STD-462. The contractor shall furnish written certification that the requirements meet MIL-STD-461 requirements. Nonconformance with the requirements specified shall constitute failure of the test.

4.4.8 Shipboard environmental test. The oven under normal operating conditions, shall be tested in accordance with MIL-STD-167/1, type I equipment. The oven shall be secured to the test machine in the same manner that it will be secured on shipboard (see 3.9.3). Failure of the oven to perform its function during and after testing, or to meet the requirements of 3.9.11 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. The oven equipment shall be preserved in accordance with the applicable requirements of MIL-B-2316.

5.1.2 Commercial. The oven equipment shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A and B packing. The oven equipment shall be packed in accordance with the applicable requirements of MIL-B-2316, preserved as specified in 5.1.

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5.2.2 Commercial packing. The oven equipment shall be packed in accordance with ASTM D 3951, preserved as specified in 5.1.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-B-2316 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The baking and roasting electric convection ovens are for use in kitchens, bakeries and galleys where combined use and time savings are required.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type and size of oven required (see 1.2).
c. When doors without windows are required (see 3.5.3).
d. When ovens are required for foreign locations and electrical characteristics required (see 3.5.6).
e. Government approved manual number to be included on data name plate (see 3.7.1).
f. When ovens are required for shipboard installations (see 3.9 through 3.9.12), and electrical characteristics required (see 3.9.5).
g. Selection of applicable level of preservation and packaging and packing (see 5.1 and 5.2).

6.2.1 Technical manuals. The requirements for technical manuals should be considered when this specification is cited on a contract. If technical manuals are required, a contract exhibit must be prepared to fully describe statement of work criteria and delivery instructions, and cite the applicable technical manual specification. The technical manuals must be acquired by separate Contract Line Item Number (CLIN) in the contract.

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

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6.4 Part number. The ovens shall be identified by a definitive part number (see 1.2.1) as shown in the following example.

M43633 - X X X X X

Electric rating
- A - 208 VAC, 60Hz, 3 phase (domestic installation only)
- B - 220 VAC, 50Hz, 1 phase (foreign installation only)
- C - 220/308 VAC, 50Hz, 3 phase - 4 wire (foreign installation only)
- D - 440 VAC, 3 phase (naval shipboard only)
- E - 230 VDC (naval shipboard only)

Use
- 1 - Naval shipboard with electromagnetic compatibility
- 2 - Naval shipboard with environmental suitability
- 3 - Naval shipboard with electromagnetic compatibility and environmental suitability
- 4 - Naval shipboard without electromagnetic compatibility and environmental suitability
- 5 - Foreign shore installation
- 6 - Domestic shore installation

Grade
- A - Oven doors with viewing window(s)
- B - Oven doors without viewing window(s)

Size
- 1 - Standard depth oven cavity
- 2 - Extra deep oven cavity

Type
- I - One oven compartment
- II - Two oven compartment

Specification number. Do not include the revision letter or activity symbol.

An item defined by inch-pounds.

6.5 Subject term (key word) listing.

Kitchen equipment

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - MD
Navy - SA, SH
Air Force - 84
DLA - GS

User activities:

Army - CE
Navy - MC

Preparing activity:

Army - GL

Project No. 7310-0711

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-43633E	Ovens, Baking & Roasting, Forced Convection, Electric

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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